CCY2001: Introduction to Cybersecurity

Lecture 5 – Introduction to Cryptography

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March 9, 2026

Lecture Outline

What is Cryptography?

What is Cryptography?

Definition

Cryptography is the science of using mathematical algorithms and coded systems to *secure communication* by protecting information from unauthorised access.

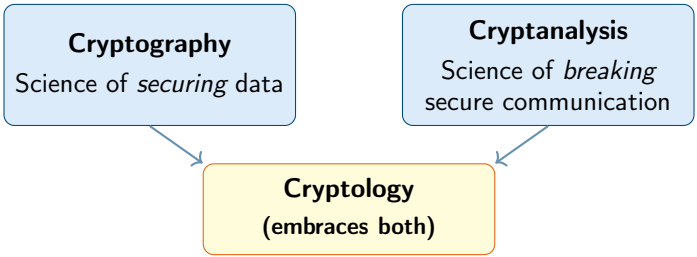
In simple words...

- Hidden / disguised / encrypted communications
- Turns readable data into unreadable form
- Only authorised parties can reverse the process

Real-life analogy

Sending a letter in a locked box. Only the person with the right key can open it and read the message.

Cryptography vs. Cryptanalysis vs. Cryptology



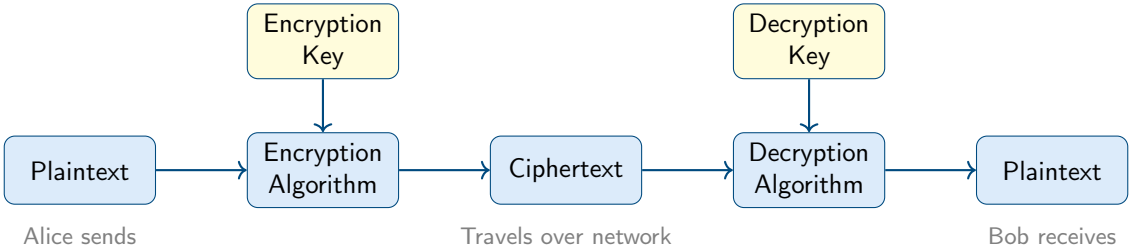
A **cryptanalyst** (attacker) tries to decode encrypted data *without* having the encryption key, using analytical reasoning, mathematics, pattern finding, and sometimes luck.

Core Terminology

Core Terminology

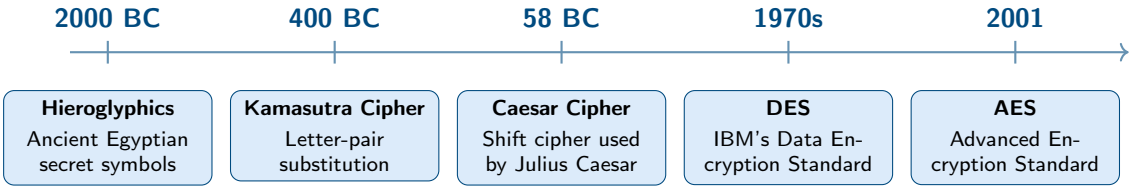
Term	Definition
Plaintext	Original, readable message (also called *cleartext*)
Ciphertext	Scrambled, unreadable form of the message
Encryption	Converting plaintext $\rightarrow$ ciphertext
Decryption	Converting ciphertext $\rightarrow$ plaintext
Cipher	Algorithm used for encryption / decryption
Key	Parameter controlling the cipher's output
Cryptosystem	Complete implementation: algorithms + infrastructure

The Cryptosystem Model



History of Cryptography

A Brief History



Hieroglyphics – The First Known Cryptography

- Oldest known use of cryptography: ≈**4000 years ago**
- Ancient Egyptians communicated using **hieroglyphs**
- Some inscriptions used non-standard or unusual symbols — a deliberate attempt to obscure meaning
- Roots of cryptography found in **Roman & Egyptian civilisations**

"As civilisations evolved, the need to communicate secretly drove the continuous evolution of cryptography."

Ancient Egyptian Symbols

Kamasutra Cipher (≈400 BC)

Kamasutra Cipher

A substitution cipher from ancient India. Letters of the alphabet are **randomly paired**; each letter is replaced by its partner.

Key: A permutation (random pairing) of the alphabet

- Originally described in the *Kama Sutra* by Vātsyāyana
- Purpose: teaching women to hide secret messages
- The key is the pairing arrangement

Example pairing (partial)

Plain	↔	Cipher
I	↔	D
N	↔	W
T	↔	R
E	↔	C
R	↔	T
S	↔	F

`INTERNET` → `DWRCTWCR`

`SOCIETY` → `FKEDCRL`

Caesar Shift Cipher

Caesar Cipher

Shift every letter in the message by a fixed number of positions in the alphabet. Julius Caesar used a **shift of 3**.

How it works (shift = 3):

$$A \to D, \quad B \to E, \quad Z \to C$$

- **Encrypt:** Plaintext $\xrightarrow{+3}$ Ciphertext
- **Decrypt:** Ciphertext $\xrightarrow{-3}$ Plaintext
- Used to protect *military messages*

Example (shift

Plaintext: HELLO
Ciphertext: KHOOR

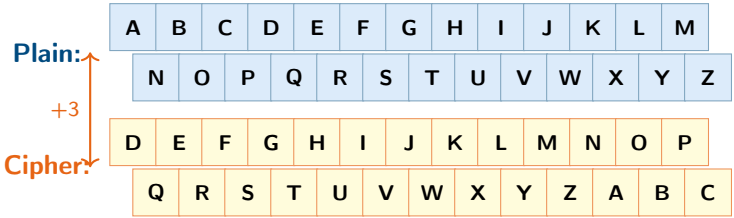
H → K
L → O
O → R

E → H
L → O

Weakness

Only 25 possible keys – easy to crack by brute force!

Caesar Cipher – Alphabet Shift Visualisation



Encrypt "ATTACK" with shift 3

A T T A C K ⟶ D W W D F N

Goals & Services of Cryptography

Goals & Services of Cryptography

Primary Goal: Secure data *at rest* and *in transit* across insecure networks.

- **Confidentiality (Secrecy)**

  Only the **intended receiver** can read the message – data stays secret even if intercepted.

- **Integrity (Anti-tampering)**

  Assures the receiver that the message has **not been altered** in transit.

- **Authentication**

  Cryptography helps establish and **prove identity** – who you are communicating with.

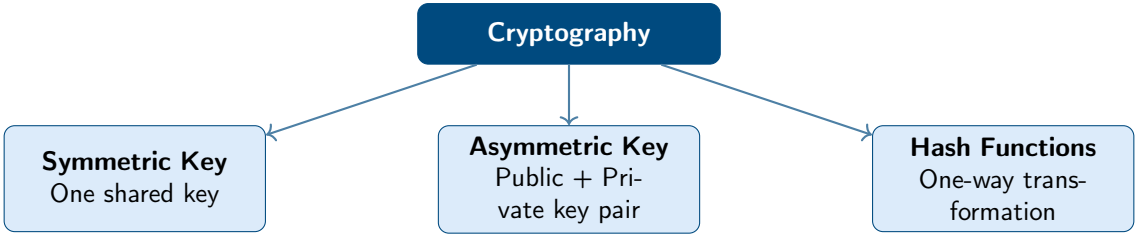
- **Non-repudiation**

  A mechanism to **prove the sender really sent** the message – they cannot deny it later.

These four services map directly to the CIA+ model you learned in earlier weeks of this course.

Types of Cryptography

Three Main Types of Cryptography

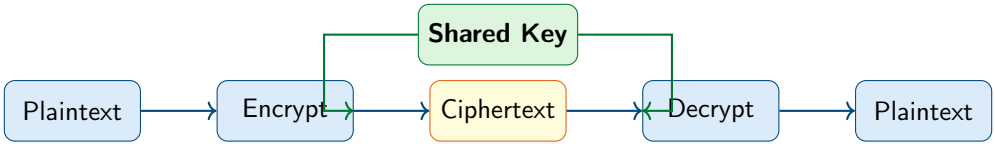


Symmetric Key Cryptography

Symmetric Key Cryptography – Overview

Symmetric Key Cryptography

Also called **Secret Key** or **Conventional** Cryptography.
The **same key** is used for both encryption and decryption.



✓ **Advantages**

- Fast and computationally efficient
- Suitable for large volumes of data

× **Drawbacks**

- Key distribution problem
- Scalability issues in large networks
- Compromised key exposes all data

Symmetric Encryption – Simple Example (Shift = 5)

Scenario: Alice sends `"HELLO"` to Bob using a Caesar-style cipher with **key = 5**

Alice – Encryption:

Plain	$\xrightarrow{+5}$	Cipher
H	→	M
E	→	J
L	→	Q
L	→	Q
O	→	T
`HELLO`		`MJQQT`

Bob – Decryption:

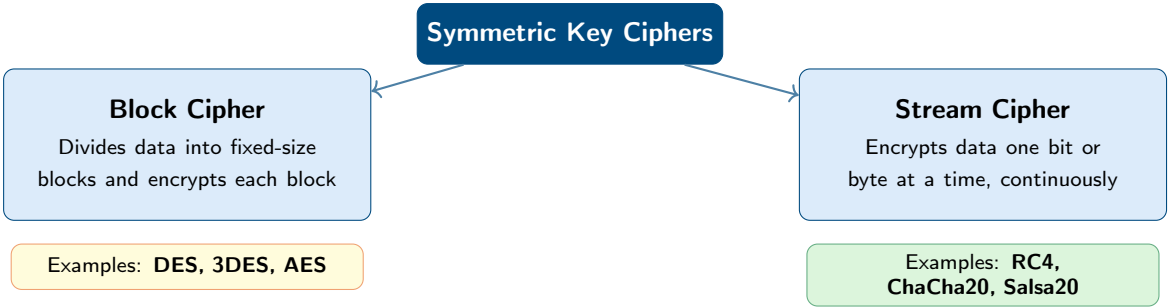
Cipher	$\xrightarrow{-5}$	Plain
M	→	H
J	→	E
Q	→	L
Q	→	L
T	→	O
`MJQQT`		`HELLO`

Both parties **must already know** the key (= 5). This is the key-distribution problem.

Symmetric Ciphers: Two Categories

Two Types of Symmetric Ciphers

Symmetric key algorithms can be divided into two fundamental categories based on **how** they process plaintext data.



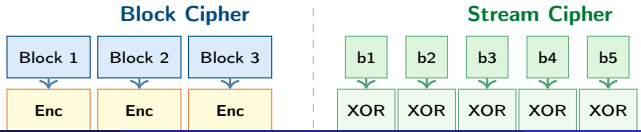
Block Cipher vs. Stream Cipher

Block Cipher

- Splits plaintext into **fixed-size blocks** (e.g. 64 or 128 bits per block)
- Encrypts the **entire block at once**
- If last block is too short $\Rightarrow$ **padding** is added
- Uses **multiple rounds** of substitution & permutation
- Generally **slower** but highly secure
- Best for: files, databases, disk encryption

Stream Cipher

- Encrypts data **one bit or byte at a time**
- Generates a **keystream** from the key and XORs it with plaintext
- **No padding** needed – works on any length
- Faster and **lower memory usage**
- Best for: real-time audio/video, network streams, mobile



Stream Cipher Examples

RC4 (Rivest Cipher 4)

- Designed by **Ron Rivest** (RSA) in **1987**
- Variable key length: 40 – 2048 bits
- Byte-oriented stream cipher
- Was widely used in: **WEP** (Wi-Fi), **WPA**, **SSL/TLS**
- Simple and very **fast** in software

RC4 Status

Deprecated. Multiple vulnerabilities found (biased keystream bytes). Prohibited in TLS since RFC 7465 (2015). **Do not use in new systems.**

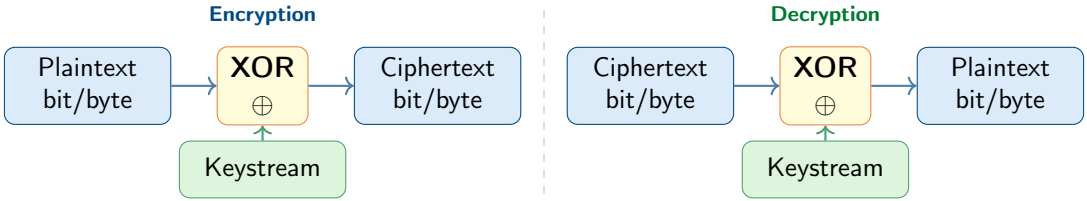
ChaCha20

- Designed by **Daniel J. Bernstein** in **2008**
- Fixed key: **256-bit**; Nonce: **96-bit**
- Based on *add-rotate-XOR* (ARX) operations
- **Modern and highly secure** – no known practical attack
- Used in: **TLS 1.3**, **QUIC**, Android, OpenSSH
- Faster than AES on devices *without* hardware acceleration

Stream Cipher – How the XOR Operation Works

The Core Idea

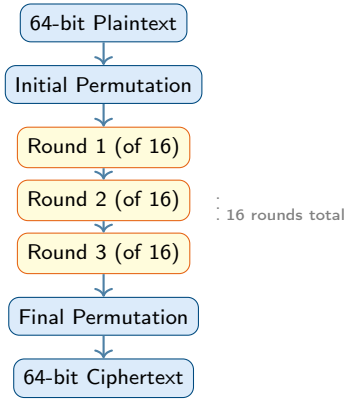
A stream cipher generates a **pseudorandom keystream** from the secret key. Each plaintext bit/byte is XOR-ed with the corresponding keystream bit/byte to produce ciphertext.



Key insight: XOR is its own inverse! Same operation for both encryption and decryption.
$P \oplus K = C$ (encrypt) $C \oplus K = P$ (decrypt)

Data Encryption Standard (DES)

- Developed by **IBM**, endorsed by NSA (1974–2002)
- Was the go-to algorithm for ≈30 years
- **Symmetric block cipher**
- Block size: **64 bits**
- Key size: **56 bits** (effective)
- Uses **16 rounds** of Feistel structure
- Includes substitution (S-boxes) and permutation



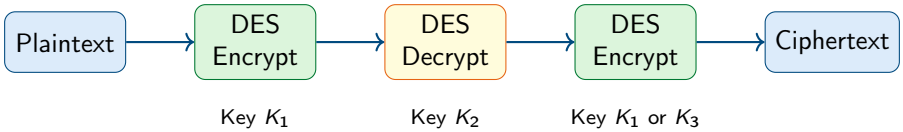
Why DES is deprecated

56-bit key $= 2^{56} \approx 72$ trillion combinations.
Modern hardware can crack DES in **hours**.

Triple DES (3DES)

Triple DES

An enhanced version of DES that applies the DES algorithm **three times** in an Encrypt–Decrypt–Encrypt (EDE) sequence.



- Effective key: **112-bit** (2-key) or **168-bit** (3-key)
- Block size still 64-bit
- More secure than DES, but **slower**

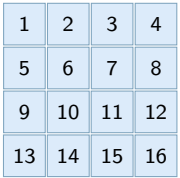
Status

NIST deprecated 3DES in 2023.
Being phased out in favour of **AES**.

Advanced Encryption Standard (AES)

AES at a Glance

Adopted by:	U.S. Government (2001)
Block size:	128 bits
Key sizes:	128 / 192 / 256 bits
Rounds:	10 / 12 / 14
Structure:	Substitution-Permutation
Status:	Current world standard



AES 4×4 State Matrix

Each cell = 1 byte; total = 128 bits

- Replaced DES as the global standard
- Extensively analysed – no practical attack known
- Used in: Wi-Fi (WPA2), TLS/HTTPS, disk encryption...

AES operations per round:
SubBytes → ShiftRows → MixColumns → AddRoundKey

Symmetric Algorithms – Comparison

Property	DES	3DES	AES
Year introduced	1977	1999	2001
Block size	64-bit	64-bit	128-bit
Key size	56-bit	112/168-bit	128/192/256-bit
Rounds	16	48	10/12/14
Speed	Fast	Slow	Fast
Security	✗ Broken	Weak	✓ Strong
Status	Deprecated	Deprecated	Current standard

Drawbacks of Symmetric Key Cryptography

The Key Distribution Problem

Both parties must share the secret key **before** communicating.
How do you securely exchange the key over an insecure network?

1. **Key Management Nightmare**
   In a network of n users, you need $\frac{n(n-1)}{2}$ unique keys!
   Example: 1000 users $\Rightarrow$ 499 500 keys
2. **Key Compromise = Total Exposure**
   If the shared key is stolen, every past and future message is exposed.
3. **No Non-repudiation**
   Since both parties share the same key, either could have created any message.

These problems motivated the invention of **Public Key (Asymmetric) Cryptography**.

Asymmetric Key Cryptography

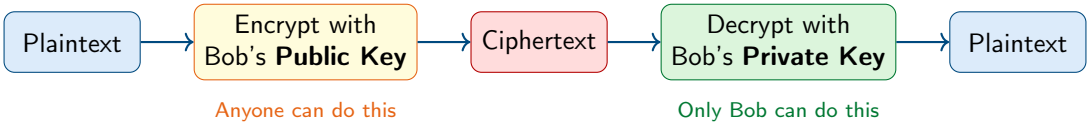
Asymmetric Key Cryptography – Overview

Asymmetric (Public Key) Cryptography

Uses **two mathematically linked keys**:

- **Public Key** – shared openly with everyone; used to *encrypt*
- **Private Key** – kept secret by the owner; used to *decrypt*

What one key encrypts, **only the other can decrypt**.



Symmetric vs. Asymmetric Encryption

Feature	Symmetric	Asymmetric
Keys used	1 shared key	2 keys (public + private)
Key exchange	Hard (insecure channel)	Easy (public key is open)
Speed	Fast	Slow
Use case	Bulk data encryption	Key exchange, signatures
Scalability	Poor (n^2 keys)	Good ($2n$ keys)
Non-repudiation	No	Yes (with private key)
Examples	AES, DES	RSA, ECC

In practice, systems use **hybrid encryption**: asymmetric to exchange a session key, then symmetric to encrypt the bulk data.
Example: TLS/HTTPS uses RSA/ECC to share an AES key.

RSA Algorithm

RSA

Published in **1977** by Rivest, Shamir, and Adleman.
First algorithm suitable for both **encryption** and **digital signing**.

Mathematical foundation:

- Easy to multiply two large primes $p \times q = n$
- Extremely hard to *factorise* n back into p and q
- This asymmetry provides security

Key sizes: 2048-bit minimum (4096-bit recommended today)

The RSA Challenge

In **1991**, RSA Laboratories released 54 semi-prime numbers and offered prizes for factorisation.

Competition ended **2007**: only **12 out of 54** factors found!

Results shaped recommendations for minimum RSA key sizes.

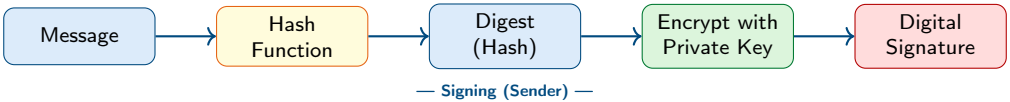
Digital Signatures

Digital Signature

A cryptographic mechanism to verify:

- **Authenticity** – the message really came from the claimed sender
- **Integrity** – the message was not changed in transit
- **Non-repudiation** – the sender cannot deny sending it

Based on **asymmetric cryptography**.



Digital Signature Standard (DSS)

- U.S. Federal standard **FIPS 186**, published by NIST in **1994**
- Specifies algorithms for generating and verifying digital signatures
- Became the U.S. government standard for *authentication* of electronic documents

Allowed algorithms under DSS:

- **DSA** – original Digital Signature Algorithm
- **RSA** – added in later revisions
- **ECDSA** – Elliptic Curve DSA (modern, efficient)

Core DSS Components

1. **Approved Algorithms:** DSA, RSA, ECDSA
2. **SHA:** Hash function produces a "digital fingerprint"
3. **PKI:** Manages public/private key pairs

DSS does not dictate *how* to sign – it specifies *which* algorithms are approved.

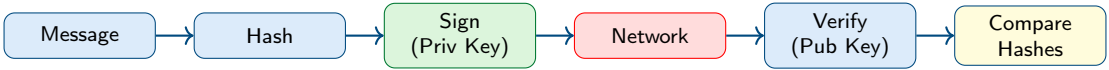
How DSS Works (RSA-Based Signatures)

Step 1 – Signing (Sender)

1. Write the message
2. Compute **hash** of the message (e.g. SHA-256)
3. **Sign** the hash with the sender's **private key**
4. This encrypted hash = **digital signature**
5. Send: message + signature

Step 2 – Verification (Receiver)

1. Receive: message + signature
2. **Recompute** the hash of the received message
3. **Verify** the signature using sender's **public key**
4. Compare both hashes
5. If they match ⇒ **valid signature**

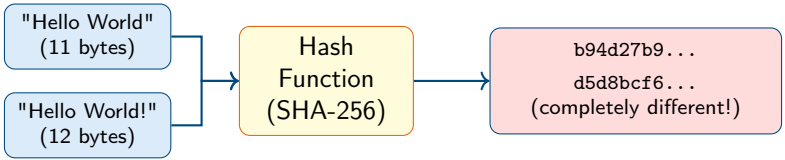


Hash Functions

Hash Functions – Overview

Hash Function

Takes **any input** (text, file, message) and produces a **fixed-size output** called a *hash* or *digest*.

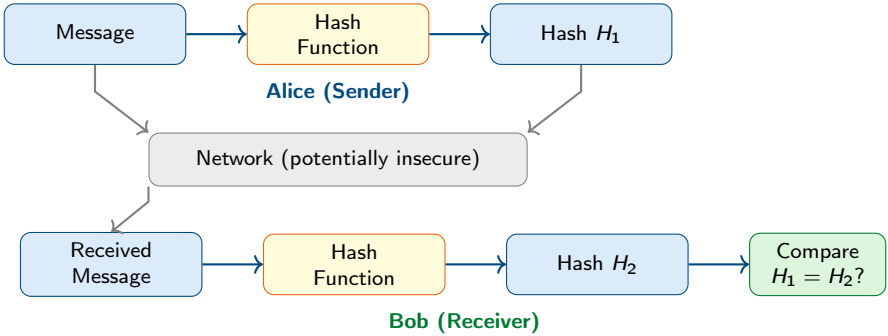


Hash Functions – Properties

Key properties:

1. **Fixed output size** regardless of input size
   (SHA-256 always outputs 256 bits)
2. **One-way (irreversible)** – you cannot reverse a hash
3. **Deterministic** – same input always gives same hash
4. **Avalanche effect** – tiny change in input $\Rightarrow$ completely different hash
5. **Collision resistant** – infeasible to find two inputs with the same hash

How Hash Functions Ensure Data Integrity



MD5 – And the Collision Problem

- **MD5**: 128-bit hash, output = 32 hex characters
- Once widely used for passwords and file integrity
- In **2005**: Wang & Yu (Shandong University) demonstrated a *collision* – two different files with the **same MD5 hash**

Collision Attack

A **collision** means two different inputs produce the same hash.
This breaks the "collision resistant" property!
An attacker could substitute a malicious file that has the same hash as the original.

MD5 Collision (2005)

`file1.dat` and `file2.dat` are **different files** yet both produce:

`MD5 = a4c0d35c...`

Checking:

```
$ md5sum file1.dat
a4c0d35c...
$ md5sum file2.dat
a4c0d35c...
```

Same hash – different files!

SHA – Secure Hash Algorithm Family

- Designed by **NSA**, published by **NIST**
- A family of hash functions: SHA-1, SHA-2, SHA-3
- Because of attacks on MD5 and SHA-1, NIST sought a new alternative
- In **October 2012**: NIST chose the **Keccak** algorithm as the new **SHA-3** standard

Algorithm	Output	Status
MD5	128-bit	Broken
SHA-1	160-bit	Broken
SHA-256	256-bit	Secure
SHA-512	512-bit	Secure
SHA-3	Variable	Current

Rule of thumb: Use **SHA-256** or **SHA-3** for any new security-sensitive application. Never use MD5 or SHA-1 for security purposes.

Summary & Review

Key Takeaways

- **Concepts**
  - Plaintext, Ciphertext, Cipher, Key
  - Encryption / Decryption
  - Cryptanalysis vs. Cryptography
- **Algorithms**
  - Symmetric: DES $\rightarrow$ 3DES $\rightarrow$ **AES**
  - Asymmetric: **RSA**, Digital Signatures, DSS
  - Hash: MD5 (broken) $\rightarrow$ **SHA-256/SHA-3**
- **History**
  - Hieroglyphs ($\approx$2000 BC)
  - Kamasutra Cipher (400 BC)
  - Caesar Cipher (shift of 3)
- **Services**
  - Confidentiality, Integrity
  - Authentication, Non-repudiation

Review Questions

1. What is the difference between **cryptography** and **cryptanalysis**?
2. Describe the **key distribution problem** in symmetric cryptography. How does asymmetric cryptography solve it?
3. Compare **DES**, **3DES**, and **AES** in terms of key size and security.
4. Explain how a **digital signature** provides non-repudiation.
5. What is a **hash collision**? Why does the MD5 collision discovered in 2005 matter for security?
6. Why is it common to use **hybrid encryption** (symmetric + asymmetric) in real-world systems like HTTPS?

Further Reading & References

- Stallings, W. (2017). *Cryptography and Network Security: Principles and Practice* (7th ed.). Pearson.
- Paar, C., & Pelzl, J. (2010). *Understanding Cryptography*. Springer.
- NIST. (2001). *Advanced Encryption Standard (FIPS 197)*.
- NIST. (2013). *Digital Signature Standard (FIPS 186-4)*.
- Anderson, R. (2020). *Security Engineering* (3rd ed.). Wiley.

- `https://csrc.nist.gov` – NIST Computer Security Resource Center
- `https://www.cryptography.com` – Introduction resources

Thank You!

Questions & Discussion

"Cryptography is the essential building block of independence for organisations on the internet."
— Whitfield Diffie

CCY2001 – Introduction to Cybersecurity
Dr. Hany Elshall